Data Analytics – Comprehensive Syllabus

Course Overview

This course provides a complete introduction to Data Analytics, focusing on data collection, cleaning, analysis, visualization, statistical methods, and business insights. Students will learn to work with real-world datasets using Python, SQL, and visualization tools to make data-driven decisions.

Duration: 10–14 Weeks (60–90 Hours)

Prerequisites:

- Basic computer knowledge
- Basic mathematics (statistics helpful but not mandatory)
- No prior programming experience required (Python basics recommended)

Module 1: Introduction to Data Analytics (Week 1)

Topics

- What is Data Analytics?
- Types of Analytics: Descriptive, Diagnostic, Predictive, Prescriptive
- Data Analytics Lifecycle
- Roles in Data Industry
- Tools used in Data Analytics
- Real-world Applications

Practical Exercises

- Understanding datasets
- Basic Excel/CSV exploration
- Simple data insights

Module 2: Excel for Data Analytics (Week 2)

Topics

- Excel Basics for Analytics
- Data Cleaning in Excel
- Sorting and Filtering
- Pivot Tables
- Charts and Graphs
- Conditional Formatting
- Basic Formulas and Functions

Practical Exercises

- Sales data analysis in Excel
- Pivot table reports
- Dashboard creation

Module 3: Python for Data Analytics (Week 3)

Topics

- Python Basics Review
- NumPy Introduction
- Pandas for Data Handling
- Series and DataFrames
- Data Import and Export
- Handling Missing Data

Practical Exercises

- Dataset loading and cleaning
- Data filtering and grouping
- Basic statistical summaries

Module 4: Data Wrangling & Cleaning (Week 4)

Topics

- Data Cleaning Techniques
- Handling Missing Values
- Removing Duplicates
- Data Transformation
- Feature Formatting
- Outlier Detection

Practical Exercises

- Real-world messy dataset cleaning
- Data preprocessing pipeline

Module 5: Exploratory Data Analysis (EDA) (Week 5)

Topics

- Understanding Data Patterns
- Summary Statistics
- Correlation Analysis
- Data Distribution
- Grouping and Aggregation

Practical Exercises

- Titanic dataset analysis
- Sales trend analysis
- Customer behavior analysis

Module 6: Data Visualization (Week 6)

Topics

- Matplotlib Basics
- Seaborn Visualization
- Types of Charts (Bar, Line, Pie, Scatter)
- Heatmaps
- Boxplots and Histograms
- Dashboard Concepts

Practical Exercises

- Sales dashboard creation
- Business performance visualization
- Trend analysis charts

Module 7: SQL for Data Analytics (Week 7)

Topics

- Introduction to Databases
- SQL Basics
- SELECT, WHERE, ORDER BY
- GROUP BY and Aggregation
- JOIN Operations
- Subqueries
- Views and Indexes

Practical Exercises

- Customer database queries
- Sales report generation
- Data extraction tasks

Module 8: Statistics for Data Analytics (Week 8)

Topics

- Descriptive Statistics

- Mean, Median, Mode
- Variance and Standard Deviation
- Probability Basics
- Correlation and Regression Basics
- Hypothesis Testing (Introduction)

Practical Exercises

- Statistical analysis using Python
- Business data interpretation
- Correlation studies

Module 9: Business Intelligence & Reporting (Week 9)

Topics

- What is Business Intelligence (BI)?
- KPI and Metrics
- Dashboard Design Principles
- Introduction to Power BI / Tableau
- Report Generation

Practical Exercises

- Interactive dashboard creation
- Business KPI reporting
- Visual storytelling

Module 10: Advanced Analytics Techniques (Week 10)

Topics

- Time Series Analysis (Basics)
- Forecasting Introduction
- Segmentation Techniques
- Anomaly Detection (Basics)

- Intro to Predictive Analytics

Practical Exercises

- Sales forecasting
- Customer segmentation
- Trend prediction

Module 11: Mini Project (Week 11)

Choose one:

- Sales Data Analysis Dashboard
- Customer Behavior Analysis
- E-commerce Data Insights
- HR Analytics Dashboard
- Stock Market Trend Analysis
- Marketing Campaign Analysis

Module 12–14: Capstone Project

End-to-End Data Analytics Project

- Problem Definition
- Data Collection
- Data Cleaning & Preparation
- Exploratory Data Analysis
- Visualization & Dashboard Creation
- Insights Generation
- Business Recommendations
- Report Preparation
- Final Presentation

Tools & Technologies

- Python
- Pandas, NumPy

- Matplotlib, Seaborn
- Jupyter Notebook / Google Colab
- SQL (MySQL / PostgreSQL)
- Microsoft Excel
- Power BI / Tableau
- Git & GitHub

Core Concepts Covered

- Data Analytics Lifecycle
- Data Cleaning & Preparation
- Exploratory Data Analysis (EDA)
- Data Visualization & Storytelling
- SQL for Data Extraction
- Statistical Analysis
- Business Intelligence (BI)
- Dashboard Development
- Basic Predictive Analytics

Assessment Scheme

Component	Weightage
Assignments	20%
Lab Exercises	20%
Quizzes	10%
Mini Project	20%
Capstone Project	30%

Learning Outcomes

By the end of this course, learners will be able to:

- Understand the complete data analytics workflow from raw data to insights.

- Clean, transform, and analyze real-world datasets using Python and Excel.
- Write SQL queries to extract and manipulate data from databases.
- Perform exploratory data analysis to identify patterns and trends.
- Create effective visualizations and interactive dashboards.
- Apply basic statistical techniques for business decision-making.
- Generate actionable insights from data for business problems.
- Build professional reports and dashboards using BI tools.
- Work on real-world analytics projects across industries like finance, healthcare, marketing, and e-commerce.